EARTHQUAKE VIBRATION DETECTOR AND ALARM

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Introduction

This project is a simple, low-cost earthquake vibration detector made using waste materials. It detects vibrations and activates a buzzer and LED to provide an immediate warning.

SCOPE of the Project

- Low-cost vibration detector
- Reuses waste cardboard
- Provides sound and light alerts
- Useful for educational and safety demonstrations
- Can be upgraded with sensors and wireless alerts

List of components

Waste cardboard – Base and support structure

Safety pin – Acts as the contact/sensing mechanism

Metal nut – Acts as the hanging vibration-sensitive weight

Buzzer – Produces the sound alarm

LED – Provides the visual warning

Battery – Supplies power to the circuit

Connecting wires – Make the electrical connections

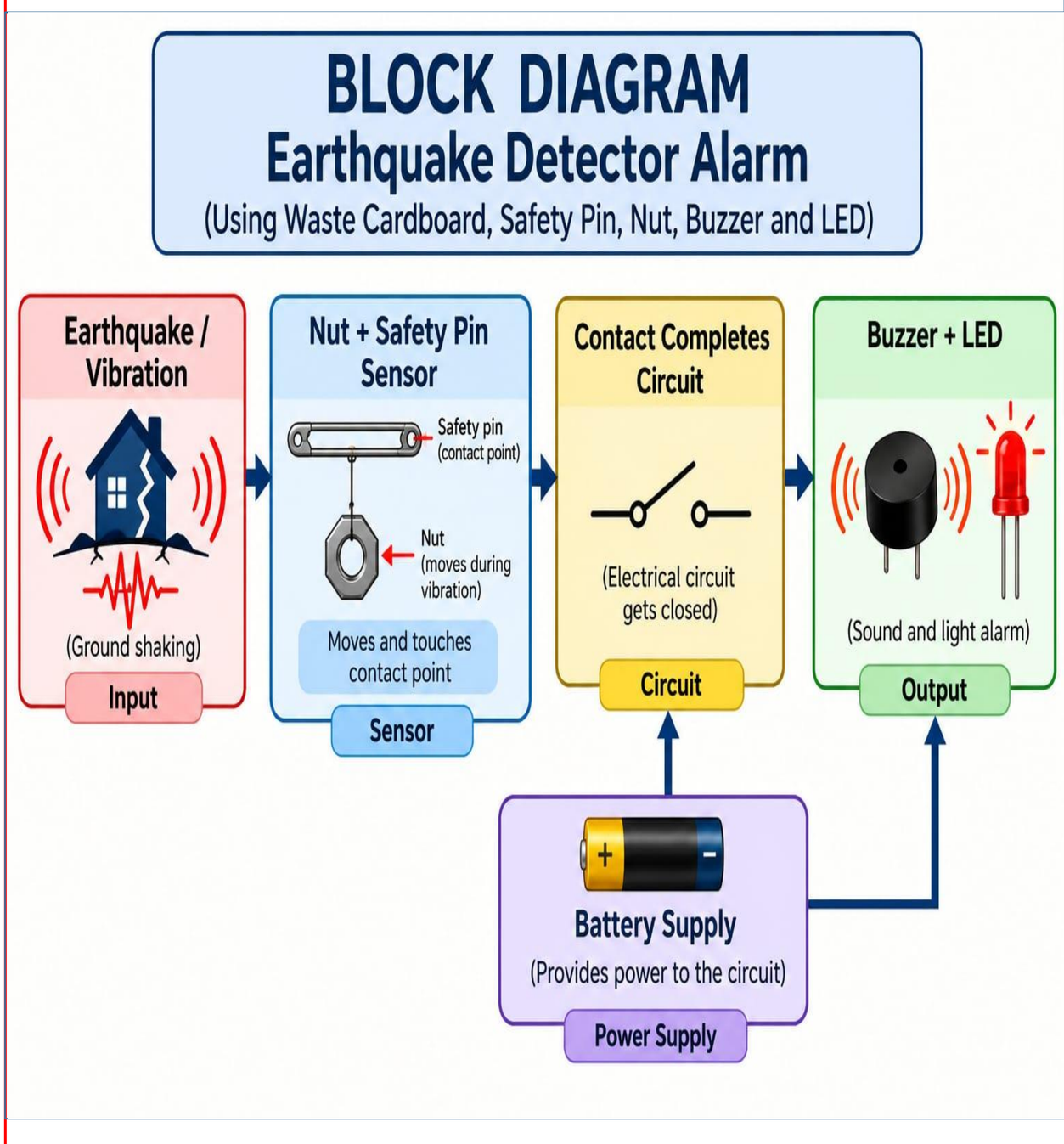
Switch – Turns the system ON/OFF

Working principle

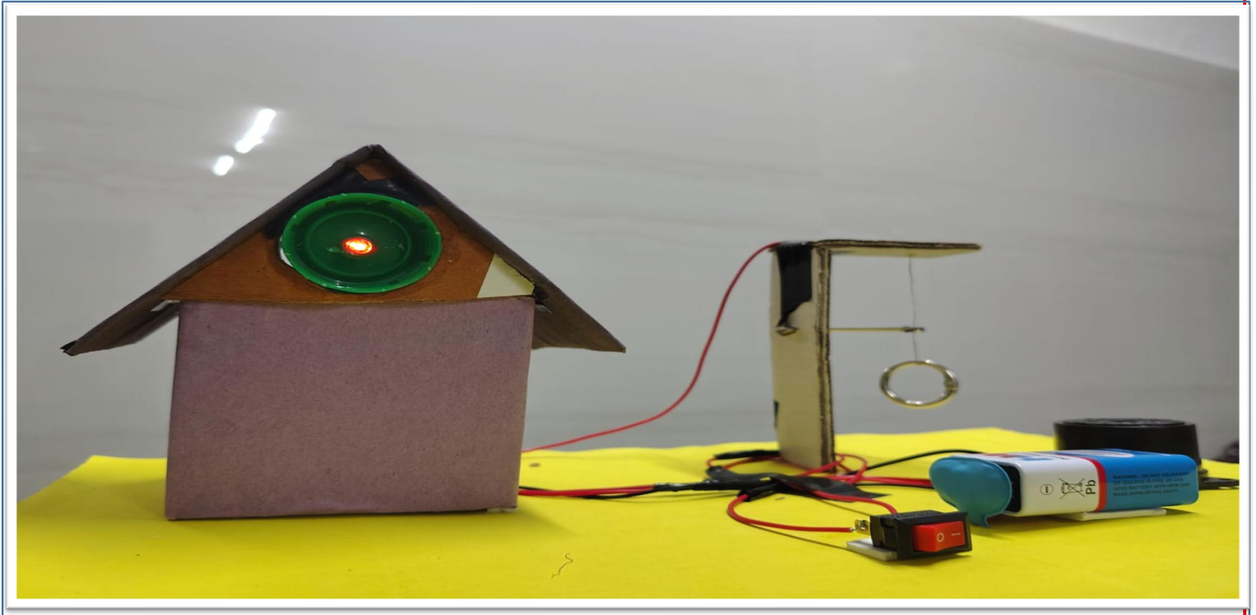
Principle **vibration-induced switching**

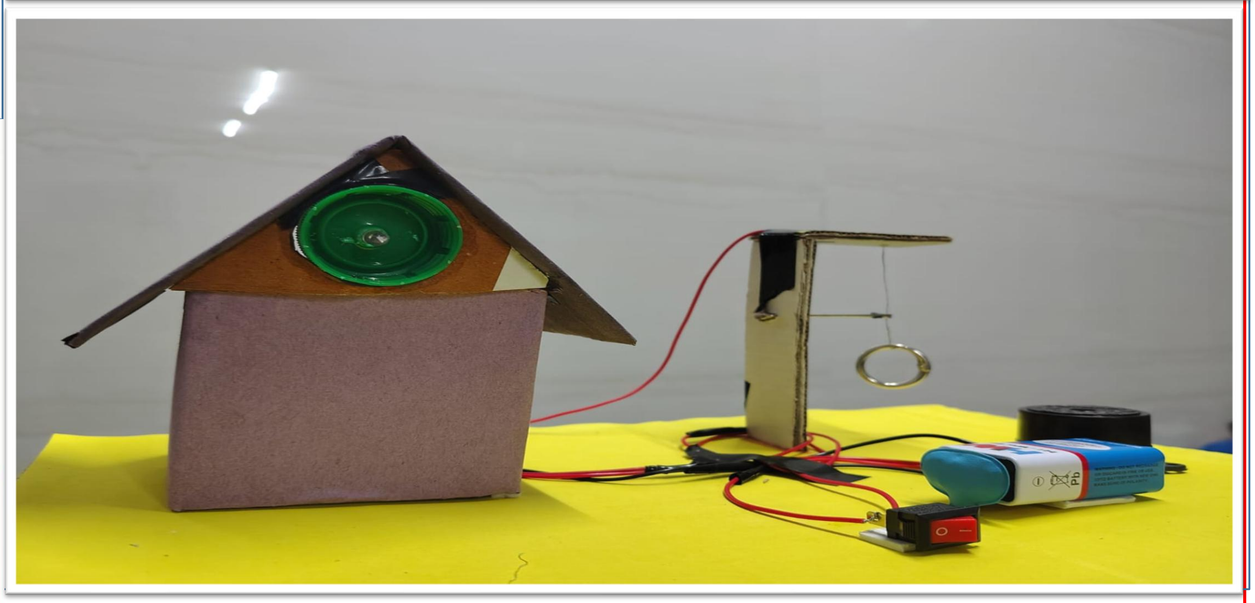
When vibration occurs, the nut moves and touches the safety-pin contact, completing the circuit. The buzzer sounds and LED glows, giving a warning.

Vibration → Nut moves → Circuit ON → Buzzer + LED ON 🚨



Photographs





Applications

- Earthquake vibration detection
- Safety alert and warning
- Educational demonstrations
- Science exhibitions
- Low-cost vibration monitoring
- Disaster awareness projects.